Question			Answer	Marks	AO	Guidance
15	(a)		vertical motion with $u\sin\theta,\ a=-g,\ s=0$ $0=u\sin\theta t-\frac{1}{2}gt^2$ $t=0,\ \frac{2u\sin\theta}{g}$	**M1**	**2.1**	*suvat* equations used with $u\sin\theta$ leading to an expression for the time of flight. Do not allow for time to the top unless subsequently doubled.
			horizontal distance R given by $t=\frac{2u\sin\theta}{g},\ R=u\cos\theta\times\frac{2u\sin\theta}{g}$	**M1**	**2.1**	horizontal motion with $u\cos\theta,\ a=0$ and their expression for t soi Allow sin/cos interchange if consistent with their vertical equation
			$R=\frac{2u^2\sin\theta\cos\theta}{g}$	**A1**	**2.1**	Convincing argument AG Note M0M1A0 for sin/cos interchange
			Alternative method Substitute $t=\frac{x}{u\cos\theta}$ to form equation of the trajectory $y=u\sin\theta\times\frac{x}{u\cos\theta}-\frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)^2$	**M1**		Allow equivalent formula quoted
			Equate y to zero and attempt to rearrange	**M1**		Allow sin/cos interchange if consistent with their horizontal equation
			$R=\frac{2u^2\sin\theta\cos\theta}{g}$	**A1**		Convincing argument AG
				[3]		

15	**(b)**		Max height H when $v_y = 0$ $0 = (u\sin\theta)^2 - 2gH$ $H = \dfrac{u^2\sin^2\theta}{2g}$ Max height exceeds range when $\dfrac{u^2\sin^2\theta}{2g} > \dfrac{2u^2\sin\theta\cos\theta}{g}$ $\tan\theta > 4$ $76.0° < \theta\,[< 90°]$	**M1** **A1** **M1** **M1** **A1**	**3.1b** **1.1b** **1.1a** **1.1a** **1.1b**	*suvat* equation(s) with $v_y = 0$ leading to an equation for H not involving t. Allow sin/cos interchange if consistent with their (a) correct expression for H Compares their H with given R Allow = used to find boundary value simplifies the inequality to an inequality for $\tan\theta$ (or equation) must be an inequality for θ. Do not penalise for omission of 90°
				[5]		